

# Science

Nature Notebook  
Nature Walks & Scouting  
Microbiology Lessons  
Microbiology Labs





## About the Course

This course includes the following topic(s): Microbiology Lessons, Microbiology Labs, Nature Notebook: Grades 9-12, Nature Walks & Scouting: Grades 9-12

**About Nature Notebook: Grades 9-12**

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

**About Nature Walks & Scouting: Grades 9-12**

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

**About Microbiology Lessons**

An elective in Life Science, Microbiology is a challenging course for the Alveary senior. Some self-direction is required.

**About Microbiology Labs**

Labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.



## Placement & Combining Tips

**Nature Notebook: Grades 9-12**

Learners may be combined and follow their own interests.

**Nature Walks & Scouting: Grades 9-12**

Learners may follow their own interests or follow the plan of their local scouting troop or natural history club.

**Microbiology Lessons**

Students should take General Biology before Microbiology. Note that there is some sensitive content in this course, and it is recommended for students in Grade 12.



## Scheduling

| GRADE | SCHEDULE INFO.                                              | BOOKS                                                                                                                                                                                                     |
|-------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9-12  | Nature Notebook: Grades 9-12<br>1+ time/week 20 min+        |                                                                                                                                                                                                           |
| 9-12  | Nature Walks & Scouting: Grades 9-12<br>1 time/week 30 min+ |                                                                                                                                                                                                           |
| 12    | Microbiology Lessons<br>5 times/week 45 min                 | A Field Guide to Bacteria<br>A Planet of Viruses<br>Growing a Revolution: Bringing Our Soil Back to Life<br>I Contain Multitudes<br>Micro Life<br>Microbe Hunters<br>The Immortal Life of Henrietta Lacks |
| 12    | Microbiology Labs<br>1 time/week 60 min                     | Illustrated Guide to Home Biology Experiments                                                                                                                                                             |

Sample Weekly View

| Day 1                                                        | Day 2                | Day 3                | Day 4                                                | Day 5                                     |
|--------------------------------------------------------------|----------------------|----------------------|------------------------------------------------------|-------------------------------------------|
| <b>Science: Microbiology</b>                                 |                      |                      |                                                      |                                           |
| Microbiology Lessons<br>Nature Walks & Scouting: Grades 9-12 | Microbiology Lessons | Microbiology Lessons | Microbiology Lessons<br>Nature Notebook: Grades 9-12 | Microbiology Lessons<br>Microbiology Labs |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

### Microbiology Lessons

- ☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.
- ☐ Bookmark or print Quick Links, as needed.
- ☐ Gather supplies for the first microbiology labs initially. You will gather additional supplies once you decide on additional labs.
- ☐ Teacher Note: Preread Chapter 15. Henrietta Lacks's children had to deal with a lot of abuse of every kind. Discuss with your student(s). The note for the student is in Week 1 of Term 2.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Science: Microbiology

#### Microbiology Lessons



A Field Guide to Bacteria



A Planet of Viruses



Growing a Revolution: Bringing Our Soil Back to Life



I Contain Multitudes



Micro Life



Microbe Hunters



The Immortal Life of Henrietta Lacks

## Microbiology Labs



Illustrated Guide to Home Biology Experiments



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

### Science: Microbiology



The Illustrated Guide to Home Biology Experiments



Student Microscope



## Quick Links

### Science: Microbiology

### Microbiology Lessons

- ∞ [Nature Journal Connection](#)
- ∞ [Alveary Bookshelf](#)
- ∞ [Illustrated Guide to Home Biology Experiments](#)
- ∞ [Seven Wonders of the Microbe World](#)
- ∞ [Microbiology from A to Z](#)
- ∞ [Microbiology Current Events](#)

Click THIS text  
or scan the QR  
code for links.



# Science: Microbiology

## How To Approach



### Introduce

- If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.
- If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.



### Read

- Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.
- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.



### Narrate

- Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.
- Use words, pictures, outlines, etc.
- If a particular idea cannot be narrated, then read or examine the text again.



### Discuss

- Consider with the teacher any thoughts, confusion, or concerns about the passage.
- If understanding is still uncertain, try rereading the passage or do some personal research on the topic.



### Connect

- Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.

# Science: Microbiology

## How To Approach Labs



### Introduce

- Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.
- Relevant concept(s) are introduced in the text.
- Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.



### Lab Procedure

- Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.
- Record all data and observations in the lab notebook.



### Analysis & Conclusions

- After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.
- If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

# Science: Microbiology

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## Term 1

### WEEK 1 45m Microbiology Lessons - Lesson 1

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 1 45m Microbiology Lessons - Lesson 2

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week, starting with the Appendices. Continue with the Introduction and read forward. Sometimes there will be practical things to do with the microscope and bacteria, but other times you will be reading to gain a greater understanding of bacteria from many different places. Sometime before next week, watch the author's video on bacteria. You may or may not agree with the author's perspective, but the point of the video is to see where some of the more recent thinking is concerning microbes.

#### → VIEW

∞ Video Link: Talk by Betsey Dexter

### WEEK 1 45m Microbiology Lessons - Lesson 3

Materials: Microbe Hunters or A Planet of Viruses

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 1 45m Microbiology Lessons - Lesson 4

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

Gather some family and friends and watch the video "Seven Wonders of the Microbe World."

#### → VIEW

∞ Video Link: Seven Wonders

### WEEK 1 45m Microbiology Lessons - Lesson 5

Materials: Illustrated Guide to Home Biology Experiments

#### → READ

Read p.3-51 at your own pace over the next few lab periods. If you already have a good working microscope of reasonable quality, you may be able to skim through the microscope chapter. If you are not that familiar with your microscope, you may benefit from reading it. You will move at your own pace through the microbiology labs in this book after completing the first 6 in order (Labs I-1 through II-3). Then, you are free to pick and choose based on your interests and/or the availability of needed supplies. The more you do, the more you will grow in your understanding of microbiology! There are several labs in the book that are general biology labs (10-12,18,20,21,23, and 24). You will skip over these and focus on the ones that require a microscope.

### WEEK 1 60m Microbiology Labs - Lesson 1

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

Continue reading p.3-51. Then, skim through the rest of the book and plot out a rough plan of which experiments you'd like to explore. Order the needed supplies. Home Scientist has a kit that contains the basic supplies



## Term 1

needed. You can add on bacterial cultures and prepared slides, if desired. If you order supplies elsewhere, you can still order the DVD with answers to the review questions in the book and high-resolution images as a separate item. As the introductory material in the book says, if you decide to do labs that require bacterial cultures or prepared slides of certain items that you cannot procure locally, you will need to order those items from Home Scientist ahead of time. They suggest that you could use the high-resolution images that come with their DVD kit instead of purchasing slides.

→ **VIEW**

∞ Link: Home Scientist

### WEEK 2 45m Microbiology Lessons - Lesson 6

Materials: I Contain Multitudes

→ **READ, NARRATE, & DISCUSS**

Read and narrate 8 pages each week.

### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 2 45m Microbiology Lessons - Lesson 7

Materials: A Field Guide to Bacteria

→ **READ, NARRATE, & DISCUSS**

Read and narrate 10 pages each week. Come up with an interesting way to present or summarize the information in this book. It could be a map, chart, diagram, or something else.

### WEEK 2 45m Microbiology Lessons - Lesson 8

Materials: Microbe Hunters or A Planet of Viruses

→ **READ, NARRATE, & DISCUSS**

Read and narrate 10 pages each week.

### WEEK 2 45m Microbiology Lessons - Lesson 9

Materials: The Immortal Life of Henrietta Lacks

→ **READ, NARRATE, & DISCUSS**

Read and narrate 10 pages each week.

### WEEK 2 45m Microbiology Lessons - Lesson 10

Materials: Illustrated Guide to Home Biology Experiments

→ **READ**

Finish reading p.3-51.

### WEEK 2 60m Microbiology Labs - Lesson 2

Materials: Illustrated Guide to Home Biology Experiments

→ **LAB DAY**

Today, you will complete LAB I-1: Using a Microscope. Gather all of the needed materials. Prepare your Lab Notebook for the experiment. Read the lab carefully before beginning and adhere to all safety requirements. Record what you do and what you observe in your Lab Notebook. Then clean up and dispose of any materials as instructed.

Complete the lab write-up in your notebook, including the review questions. After you have finished your work, use the answers in the PDF





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that came with your supplies and see if there was anything significant that you did not understand. Use your Biology textbook from Grade 10 and the internet to learn more or aid in understanding terms and concepts.

You will complete your lab report following the book's guidelines. Choose 6 throughout the year to complete as full lab reports. Use what you have learned about writing lab reports. Use the guidelines for the rest of your reports.

### WEEK 3 45m Microbiology Lessons - Lesson 11

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 3 45m Microbiology Lessons - Lesson 12

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 3 45m Microbiology Lessons - Lesson 13

Materials: Microbe Hunters or A Planet of Viruses

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 3 45m Microbiology Lessons - Lesson 14

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 3 45m Microbiology Lessons - Lesson 15

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 3 60m Microbiology Labs - Lesson 3

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 4 45m Microbiology Lessons - Lesson 16

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 4 45m Microbiology Lessons - Lesson 17

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

# Science: Microbiology

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### WEEK 4 45m Microbiology Lessons - Lesson 18

Materials: Microbe Hunters or A Planet of Viruses

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

Gather some family and friends and watch the video "How to Identify Microbes."

#### → VIEW

∞ Video Link: How to Identify Microbes

### WEEK 4 45m Microbiology Lessons - Lesson 19

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 4 45m Microbiology Lessons - Lesson 20

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 4 60m Microbiology Labs - Lesson 4

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 5 45m Microbiology Lessons - Lesson 21

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 5 45m Microbiology Lessons - Lesson 22

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 5 45m Microbiology Lessons - Lesson 23

Materials: Microbe Hunters or A Planet of Viruses

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 5 45m Microbiology Lessons - Lesson 24

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 5 45m Microbiology Lessons - Lesson 25

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

# Science: Microbiology

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## Term 1

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>WEEK 5</b> <b>60m Microbiology Labs - Lesson 5</b><br><div>  Materials: Illustrated Guide to Home Biology Experiments         </div><br><b>→ LAB DAY</b>                                                           |                                                                                |
| <b>WEEK 6</b> <b>45m Microbiology Lessons - Lesson 26</b><br><div>  Materials: I Contain Multitudes         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 8 pages each week.                    | <b>• PLAN WEEKLY</b><br>nature walk - record observations<br>science free read |
| <b>WEEK 6</b> <b>45m Microbiology Lessons - Lesson 27</b><br><div>  Materials: A Field Guide to Bacteria         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week.              |                                                                                |
| <b>WEEK 6</b> <b>45m Microbiology Lessons - Lesson 28</b><br><div>  Materials: Microbe Hunters or A Planet of Viruses         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week. |                                                                                |
| <b>WEEK 6</b> <b>45m Microbiology Lessons - Lesson 29</b><br><div>  Materials: The Immortal Life of Henrietta Lacks         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week.   |                                                                                |
| <b>WEEK 6</b> <b>45m Microbiology Lessons - Lesson 30</b><br><div>  Materials: Illustrated Guide to Home Biology Experiments         </div><br><b>→ LAB DAY</b>                                                       |                                                                                |
| <b>WEEK 6</b> <b>60m Microbiology Labs - Lesson 6</b><br><div>  Materials: Illustrated Guide to Home Biology Experiments         </div><br><b>→ LAB DAY</b>                                                           |                                                                                |
| <b>WEEK 7</b> <b>45m Microbiology Lessons - Lesson 31</b><br><div>  Materials: I Contain Multitudes         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 8 pages each week.                    | <b>• PLAN WEEKLY</b><br>nature walk - record observations<br>science free read |
| <b>WEEK 7</b> <b>45m Microbiology Lessons - Lesson 32</b><br><div>  Materials: A Field Guide to Bacteria         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week.              |                                                                                |

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## Term 1

### WEEK 7 ☐ 45m Microbiology Lessons - Lesson 33

☐ Materials: Microbe Hunters or A Planet of Viruses

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 7 ☐ 45m Microbiology Lessons - Lesson 34

☐ Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 7 ☐ 45m Microbiology Lessons - Lesson 35

☐ Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 7 ☐ 60m Microbiology Labs - Lesson 7

☐ Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 8 ☐ 45m Microbiology Lessons - Lesson 36

☐ Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 8 ☐ 45m Microbiology Lessons - Lesson 37

☐ Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 8 ☐ 45m Microbiology Lessons - Lesson 38

☐ Materials: Microbe Hunters or A Planet of Viruses

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 8 ☐ 45m Microbiology Lessons - Lesson 39

☐ Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 8 ☐ 45m Microbiology Lessons - Lesson 40

☐ Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

# Science: Microbiology

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|                                                                                                                                                                                                                       |                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>WEEK 8</b> <b>60m Microbiology Labs - Lesson 8</b><br><div>  Materials: Illustrated Guide to Home Biology Experiments         </div><br><b>→ LAB DAY</b>                                                           |                                                                                |
| <b>WEEK 9</b> <b>45m Microbiology Lessons - Lesson 41</b><br><div>  Materials: I Contain Multitudes         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 8 pages each week.                    | <b>• PLAN WEEKLY</b><br>nature walk - record observations<br>science free read |
| <b>WEEK 9</b> <b>45m Microbiology Lessons - Lesson 42</b><br><div>  Materials: A Field Guide to Bacteria         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week.              |                                                                                |
| <b>WEEK 9</b> <b>45m Microbiology Lessons - Lesson 43</b><br><div>  Materials: Microbe Hunters or A Planet of Viruses         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week. |                                                                                |
| <b>WEEK 9</b> <b>45m Microbiology Lessons - Lesson 44</b><br><div>  Materials: The Immortal Life of Henrietta Lacks         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week.   |                                                                                |
| <b>WEEK 9</b> <b>45m Microbiology Lessons - Lesson 45</b><br><div>  Materials: Illustrated Guide to Home Biology Experiments         </div><br><b>→ LAB DAY</b>                                                       |                                                                                |
| <b>WEEK 9</b> <b>60m Microbiology Labs - Lesson 9</b><br><div>  Materials: Illustrated Guide to Home Biology Experiments         </div><br><b>→ LAB DAY</b>                                                           |                                                                                |
| <b>WEEK 10</b> <b>45m Microbiology Lessons - Lesson 46</b><br><div>  Materials: I Contain Multitudes         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 8 pages each week.                   | <b>• PLAN WEEKLY</b><br>nature walk - record observations<br>science free read |
| <b>WEEK 10</b> <b>45m Microbiology Lessons - Lesson 47</b><br><div>  Materials: A Field Guide to Bacteria         </div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week.             |                                                                                |

# Science: Microbiology

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## Term 1

### WEEK 10 ☐ 45m Microbiology Lessons - Lesson 48

☐ Materials: Microbe Hunters or A Planet of Viruses

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 10 ☐ 45m Microbiology Lessons - Lesson 49

☐ Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 10 ☐ 45m Microbiology Lessons - Lesson 50

☐ Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 10 ☐ 60m Microbiology Labs - Lesson 10

☐ Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 11 ☐ 45m Microbiology Lessons - Lesson 51

☐ Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 11 ☐ 45m Microbiology Lessons - Lesson 52

☐ Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 11 ☐ 45m Microbiology Lessons - Lesson 53

☐ Materials: Microbe Hunters or A Planet of Viruses

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 11 ☐ 45m Microbiology Lessons - Lesson 54

☐ Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 11 ☐ 45m Microbiology Lessons - Lesson 55

☐ Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

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## Term 1

### WEEK 11 60m Microbiology Labs - Lesson 11

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 12 45m Microbiology Lessons - Lesson 56

*Exam Week*

→ EXAMS

Answer question(s) related to course.

### WEEK 12 45m Microbiology Lessons - Lesson 57

*Exam Week*

→ EXAMS

Answer question(s) related to course.

### WEEK 12 45m Microbiology Lessons - Lesson 58

*Exam Week*

→ EXAMS

Answer question(s) related to course.

### WEEK 12 45m Microbiology Lessons - Lesson 59

*Exam Week*

→ EXAMS

Answer question(s) related to course.

### WEEK 12 45m Microbiology Lessons - Lesson 60

*Exam Week*

→ EXAMS

Answer question(s) related to course.

### WEEK 12 60m Microbiology Labs - Lesson 12

*Exam Week*

→ EXAMS

Answer question(s) related to course.

# Science: Microbiology

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## Term 2

### WEEK 1 45m Microbiology Lessons - Lesson 61

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Continue to read and narrate 8 pages each week. Challenge yourself to come up with a new way of narrating. Ask questions of the text and seek to find the answers. Discuss the ideas with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 1 45m Microbiology Lessons - Lesson 62

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 1 45m Microbiology Lessons - Lesson 63

Materials: Microbe Hunters or Micro Life

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR skim-read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

Gather some family and friends and watch the video "Wonders of the Microbial World."

#### → VIEW

∞ Video Link: Wonders of the Microbial World

### WEEK 1 45m Microbiology Lessons - Lesson 64

Materials: The Immortal Life of Henrietta Lacks

NOTE: Sensitive content

Chapter 15 contains disturbing accounts of child physical and sexual abuse. Discuss with your teacher before reading.

### WEEK 1 45m Microbiology Lessons - Lesson 65

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

Continue lab work, writing up your report after completing each lab.

### WEEK 1 60m Microbiology Labs - Lesson 13

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

Continue lab work, and write up your report after completing each lab.

### WEEK 2 45m Microbiology Lessons - Lesson 66

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 2 45m Microbiology Lessons - Lesson 67

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.



# Science: Microbiology

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## Term 2

### WEEK 2 45m Microbiology Lessons - Lesson 68

Materials: Microbe Hunters or Micro Life

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

### WEEK 2 45m Microbiology Lessons - Lesson 69

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 2 45m Microbiology Lessons - Lesson 70

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 2 60m Microbiology Labs - Lesson 14

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 3 45m Microbiology Lessons - Lesson 71

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 3 45m Microbiology Lessons - Lesson 72

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 3 45m Microbiology Lessons - Lesson 73

Materials: Microbe Hunters or Micro Life

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

### WEEK 3 45m Microbiology Lessons - Lesson 74

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 3 45m Microbiology Lessons - Lesson 75

Materials: Illustrated Guide to Home Biology Experiments

# Science: Microbiology

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## Term 2

→ LAB DAY

### WEEK 3 ☐ 60m Microbiology Labs - Lesson 15

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 4 ☐ 45m Microbiology Lessons - Lesson 76

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 4 ☐ 45m Microbiology Lessons - Lesson 77

☐ Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 4 ☐ 45m Microbiology Lessons - Lesson 78

☐ Materials: Microbe Hunters or Micro Life

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

### WEEK 4 ☐ 45m Microbiology Lessons - Lesson 79

☐ Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 4 ☐ 45m Microbiology Lessons - Lesson 80

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 4 ☐ 60m Microbiology Labs - Lesson 16

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 5 ☐ 45m Microbiology Lessons - Lesson 81

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 5 ☐ 45m Microbiology Lessons - Lesson 82

☐ Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages a week.

# Science: Microbiology

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## Term 2

|                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>WEEK 5</b> <b>45m Microbiology Lessons - Lesson 83</b></p> <p> Materials: Microbe Hunters or Micro Life</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.</p> | <p>Gather some family and friends and watch something interesting from this YouTube channel.</p> <p>→ <b>VIEW</b><br/>∞ <b>Video Link:</b> Journey to the Microcosmos</p> |
| <p><b>WEEK 5</b> <b>45m Microbiology Lessons - Lesson 84</b></p> <p> Materials: The Immortal Life of Henrietta Lacks</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week. Discuss with your teacher.</p>                                                                                            |                                                                                                                                                                           |
| <p><b>WEEK 5</b> <b>45m Microbiology Lessons - Lesson 85</b></p> <p> Materials: Illustrated Guide to Home Biology Experiments</p> <p>→ <b>LAB DAY</b></p>                                                                                                                                                                            |                                                                                                                                                                           |
| <p><b>WEEK 5</b> <b>60m Microbiology Labs - Lesson 17</b></p> <p> Materials: Illustrated Guide to Home Biology Experiments</p> <p>→ <b>LAB DAY</b></p>                                                                                                                                                                               |                                                                                                                                                                           |
| <p><b>WEEK 6</b> <b>45m Microbiology Lessons - Lesson 86</b></p> <p> Materials: I Contain Multitudes</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 8 pages each week. Discuss with others.</p>                                                                                                                   | <p>• <b>PLAN WEEKLY</b></p> <ul style="list-style-type: none"> <li> nature walk - record observations</li> <li> science free read</li> </ul>                              |
| <p><b>WEEK 6</b> <b>45m Microbiology Lessons - Lesson 87</b></p> <p> Materials: A Field Guide to Bacteria</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week.</p>                                                                                                                                  |                                                                                                                                                                           |
| <p><b>WEEK 6</b> <b>45m Microbiology Lessons - Lesson 88</b></p> <p> Materials: Microbe Hunters or Micro Life</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.</p> |                                                                                                                                                                           |
| <p><b>WEEK 6</b> <b>45m Microbiology Lessons - Lesson 89</b></p> <p> Materials: The Immortal Life of Henrietta Lacks</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week. Discuss with your teacher.</p>                                                                                            |                                                                                                                                                                           |
| <p><b>WEEK 6</b> <b>45m Microbiology Lessons - Lesson 90</b></p> <p> Materials: Illustrated Guide to Home Biology Experiments</p>                                                                                                                                                                                                    |                                                                                                                                                                           |

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## Term 2

→ LAB DAY

### WEEK 6 60m Microbiology Labs - Lesson 18

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 7 45m Microbiology Lessons - Lesson 91

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 7 45m Microbiology Lessons - Lesson 92

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 7 45m Microbiology Lessons - Lesson 93

Materials: Microbe Hunters or Micro Life

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

### WEEK 7 45m Microbiology Lessons - Lesson 94

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 7 45m Microbiology Lessons - Lesson 95

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 7 60m Microbiology Labs - Lesson 19

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 8 45m Microbiology Lessons - Lesson 96

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 8 45m Microbiology Lessons - Lesson 97

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

# Science: Microbiology

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## Term 2

### WEEK 8 45m Microbiology Lessons - Lesson 98

Materials: Microbe Hunters or Micro Life

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

### WEEK 8 45m Microbiology Lessons - Lesson 99

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 8 45m Microbiology Lessons - Lesson 100

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 8 60m Microbiology Labs - Lesson 20

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 9 45m Microbiology Lessons - Lesson 101

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 9 45m Microbiology Lessons - Lesson 102

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 9 45m Microbiology Lessons - Lesson 103

Materials: Microbe Hunters or Micro Life

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

Gather some family and friends and watch something interesting from this YouTube channel.

#### → VIEW

∞ **Video Link:** Journey to the Microcosmos

### WEEK 9 45m Microbiology Lessons - Lesson 104

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 9 45m Microbiology Lessons - Lesson 105

Materials: Illustrated Guide to Home Biology Experiments

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## Term 2

→ LAB DAY

### WEEK 9 60m Microbiology Labs - Lesson 21

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 10 45m Microbiology Lessons - Lesson 106

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 10 45m Microbiology Lessons - Lesson 107

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 10 45m Microbiology Lessons - Lesson 108

Materials: Microbe Hunters or Micro Life

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

### WEEK 10 45m Microbiology Lessons - Lesson 109

Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 10 45m Microbiology Lessons - Lesson 110

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 10 60m Microbiology Labs - Lesson 22

Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 11 45m Microbiology Lessons - Lesson 111

Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 11 45m Microbiology Lessons - Lesson 112

Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

# Science: Microbiology

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## Term 2

### WEEK 11 45m Microbiology Lessons - Lesson 113

Materials: Microbe Hunters or Micro Life

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters, OR read and look at the pictures on ~10 two-page spreads from Micro Life and sketch some of the images.

### WEEK 11 45m Microbiology Lessons - Lesson 114

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 11 45m Microbiology Lessons - Lesson 115

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 11 60m Microbiology Labs - Lesson 23

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 12 45m Microbiology Lessons - Lesson 116

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

### WEEK 12 45m Microbiology Lessons - Lesson 117

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

### WEEK 12 45m Microbiology Lessons - Lesson 118

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

### WEEK 12 45m Microbiology Lessons - Lesson 119

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

### WEEK 12 45m Microbiology Lessons - Lesson 120

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

# Science: Microbiology

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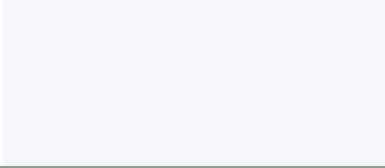
## Term 2

WEEK 12 60m Microbiology Labs - Lesson 24

*Exam Week*

→ EXAMS

Answer question(s) related to course.





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## Term 3

|        |                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                           |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WEEK 1 | <p><b>45m Microbiology Lessons - Lesson 121</b></p> <p>Materials: I Contain Multitudes</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Continue to read and narrate 8 pages each week. Challenge yourself to come up with a new way of narrating. Ask questions of the text and seek to find the answers. Discuss the ideas with others.</p>                                                               | <p>• <b>PLAN WEEKLY</b></p> <p>nature walk - record observations</p> <p>science free read</p>                                                                             |
| WEEK 1 | <p><b>45m Microbiology Lessons - Lesson 122</b></p> <p>Materials: A Field Guide to Bacteria</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week.</p>                                                                                                                                                                                                                       |                                                                                                                                                                           |
| WEEK 1 | <p><b>45m Microbiology Lessons - Lesson 123</b></p> <p>Materials: Microbe Hunters OR Growing a Revolution</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.</p> | <p>Gather some family and friends and watch something interesting from this YouTube channel.</p> <p>→ <b>VIEW</b><br/>∞ <b>Video Link:</b> Journey to the Microcosmos</p> |
| WEEK 1 | <p><b>45m Microbiology Lessons - Lesson 124</b></p> <p>Materials: The Immortal Life of Henrietta Lacks</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week. Discuss with your teacher.</p>                                                                                                                                                                                 |                                                                                                                                                                           |
| WEEK 1 | <p><b>45m Microbiology Lessons - Lesson 125</b></p> <p>Materials: Illustrated Guide to Home Biology Experiments</p> <p>→ <b>LAB DAY</b><br/>Continue lab work, writing up your report after completing each lab.</p>                                                                                                                                                                                        |                                                                                                                                                                           |
| WEEK 1 | <p><b>60m Microbiology Labs - Lesson 25</b></p> <p>Materials: Illustrated Guide to Home Biology Experiments</p> <p>→ <b>LAB DAY</b><br/>Continue lab work, writing up your report after completing each lab.</p>                                                                                                                                                                                            |                                                                                                                                                                           |
| WEEK 2 | <p><b>45m Microbiology Lessons - Lesson 126</b></p> <p>Materials: I Contain Multitudes</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 8 pages each week. Discuss with others.</p>                                                                                                                                                                                                        | <p>• <b>PLAN WEEKLY</b></p> <p>nature walk - record observations</p> <p>science free read</p>                                                                             |
| WEEK 2 | <p><b>45m Microbiology Lessons - Lesson 127</b></p> <p>Materials: A Field Guide to Bacteria</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week.</p>                                                                                                                                                                                                                       |                                                                                                                                                                           |



## Term 3

### WEEK 2 45m Microbiology Lessons - Lesson 128

Materials: Microbe Hunters OR Growing a Revolution

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

### WEEK 2 45m Microbiology Lessons - Lesson 129

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 2 45m Microbiology Lessons - Lesson 130

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 2 60m Microbiology Labs - Lesson 26

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 3 45m Microbiology Lessons - Lesson 131

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 3 45m Microbiology Lessons - Lesson 132

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 3 45m Microbiology Lessons - Lesson 133

Materials: Microbe Hunters OR Growing a Revolution

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

### WEEK 3 45m Microbiology Lessons - Lesson 134

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

# Science: Microbiology

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## Term 3

### WEEK 3 ☐ 45m Microbiology Lessons - Lesson 135

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 3 ☐ 60m Microbiology Labs - Lesson 27

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 4 ☐ 45m Microbiology Lessons - Lesson 136

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 4 ☐ 45m Microbiology Lessons - Lesson 137

☐ Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 4 ☐ 45m Microbiology Lessons - Lesson 138

☐ Materials: Microbe Hunters OR Growing a Revolution

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

### WEEK 4 ☐ 45m Microbiology Lessons - Lesson 139

☐ Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 4 ☐ 45m Microbiology Lessons - Lesson 140

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 4 ☐ 60m Microbiology Labs - Lesson 28

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 5 ☐ 45m Microbiology Lessons - Lesson 141

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

# Science: Microbiology

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## Term 3

### WEEK 5 45m Microbiology Lessons - Lesson 142

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 5 45m Microbiology Lessons - Lesson 143

Materials: Microbe Hunters OR Growing a Revolution

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

Gather some family and friends and watch something interesting from this YouTube channel.

#### → VIEW

∞ Video Link: Journey to the Microcosmos

### WEEK 5 45m Microbiology Lessons - Lesson 144

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 5 45m Microbiology Lessons - Lesson 145

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 5 60m Microbiology Labs - Lesson 29

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 6 45m Microbiology Lessons - Lesson 146

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 6 45m Microbiology Lessons - Lesson 147

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 6 45m Microbiology Lessons - Lesson 148

Materials: Microbe Hunters OR Growing a Revolution

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

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## Term 3

### WEEK 6 45m Microbiology Lessons - Lesson 149

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 6 45m Microbiology Lessons - Lesson 150

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 6 60m Microbiology Labs - Lesson 30

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 7 45m Microbiology Lessons - Lesson 151

Materials: I Contain Multitudes

#### → READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

nature walk - record observations

science free read

### WEEK 7 45m Microbiology Lessons - Lesson 152

Materials: A Field Guide to Bacteria

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 7 45m Microbiology Lessons - Lesson 153

Materials: Microbe Hunters OR Growing a Revolution

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

### WEEK 7 45m Microbiology Lessons - Lesson 154

Materials: The Immortal Life of Henrietta Lacks

#### → READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 7 45m Microbiology Lessons - Lesson 155

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

### WEEK 7 60m Microbiology Labs - Lesson 31

Materials: Illustrated Guide to Home Biology Experiments

#### → LAB DAY

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## Term 3

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>WEEK 8</b> <input type="checkbox"/> <b>45m Microbiology Lessons - Lesson 156</b></p> <p><input type="checkbox"/> Materials: I Contain Multitudes</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 8 pages each week. Discuss with others.</p>                                                                                                                                                                                                        | <p>• <b>PLAN WEEKLY</b></p> <p><input type="checkbox"/> nature walk - record observations</p> <p><input type="checkbox"/> science free read</p> |
| <p><b>WEEK 8</b> <input type="checkbox"/> <b>45m Microbiology Lessons - Lesson 157</b></p> <p><input type="checkbox"/> Materials: A Field Guide to Bacteria</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week.</p>                                                                                                                                                                                                                       |                                                                                                                                                 |
| <p><b>WEEK 8</b> <input type="checkbox"/> <b>45m Microbiology Lessons - Lesson 158</b></p> <p><input type="checkbox"/> Materials: Microbe Hunters OR Growing a Revolution</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.</p> |                                                                                                                                                 |
| <p><b>WEEK 8</b> <input type="checkbox"/> <b>45m Microbiology Lessons - Lesson 159</b></p> <p><input type="checkbox"/> Materials: The Immortal Life of Henrietta Lacks</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week. Discuss with your teacher.</p>                                                                                                                                                                                 |                                                                                                                                                 |
| <p><b>WEEK 8</b> <input type="checkbox"/> <b>45m Microbiology Lessons - Lesson 160</b></p> <p><input type="checkbox"/> Materials: Illustrated Guide to Home Biology Experiments</p> <p>→ <b>LAB DAY</b></p>                                                                                                                                                                                                                                                                 |                                                                                                                                                 |
| <p><b>WEEK 8</b> <input type="checkbox"/> <b>60m Microbiology Labs - Lesson 32</b></p> <p><input type="checkbox"/> Materials: Illustrated Guide to Home Biology Experiments</p> <p>→ <b>LAB DAY</b></p>                                                                                                                                                                                                                                                                     |                                                                                                                                                 |
| <p><b>WEEK 9</b> <input type="checkbox"/> <b>45m Microbiology Lessons - Lesson 161</b></p> <p><input type="checkbox"/> Materials: I Contain Multitudes</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 8 pages each week. Discuss with others.</p>                                                                                                                                                                                                        | <p>• <b>PLAN WEEKLY</b></p> <p><input type="checkbox"/> nature walk - record observations</p> <p><input type="checkbox"/> science free read</p> |
| <p><b>WEEK 9</b> <input type="checkbox"/> <b>45m Microbiology Lessons - Lesson 162</b></p> <p><input type="checkbox"/> Materials: A Field Guide to Bacteria</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week.</p>                                                                                                                                                                                                                       |                                                                                                                                                 |
| <p><b>WEEK 9</b> <input type="checkbox"/> <b>45m Microbiology Lessons - Lesson 163</b></p> <p><input type="checkbox"/> Materials: Microbe Hunters OR Growing a Revolution</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>Read and narrate 10 pages each week from Microbe Hunters OR about a</p>                                                                                                                                                                          | <p>Gather some family and friends and watch something interesting from this YouTube channel.</p> <p>→ <b>VIEW</b></p>                           |

# Science: Microbiology

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## Term 3

|                |                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|                | chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.                                                                                                                                                                                                                               | ∞ <b>Video Link:</b> Journey to the Microcosmos                                                      |
| <b>WEEK 9</b>  | <b>45m Microbiology Lessons - Lesson 164</b><br><div>Materials: The Immortal Life of Henrietta Lacks</div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week. Discuss with your teacher.                                                                                                                                                                                 |                                                                                                      |
| <b>WEEK 9</b>  | <b>45m Microbiology Lessons - Lesson 165</b><br><div>Materials: Illustrated Guide to Home Biology Experiments</div><br><b>→ LAB DAY</b>                                                                                                                                                                                                                                                                |                                                                                                      |
| <b>WEEK 9</b>  | <b>60m Microbiology Labs - Lesson 33</b><br><div>Materials: Illustrated Guide to Home Biology Experiments</div><br><b>→ LAB DAY</b>                                                                                                                                                                                                                                                                    |                                                                                                      |
| <b>WEEK 10</b> | <b>45m Microbiology Lessons - Lesson 166</b><br><div>Materials: I Contain Multitudes</div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 8 pages each week. Discuss with others.                                                                                                                                                                                                        | <b>• PLAN WEEKLY</b><br><div>nature walk - record observations</div><br><div>science free read</div> |
| <b>WEEK 10</b> | <b>45m Microbiology Lessons - Lesson 167</b><br><div>Materials: A Field Guide to Bacteria</div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week.                                                                                                                                                                                                                       |                                                                                                      |
| <b>WEEK 10</b> | <b>45m Microbiology Lessons - Lesson 168</b><br><div>Materials: Microbe Hunters OR Growing a Revolution</div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading. |                                                                                                      |
| <b>WEEK 10</b> | <b>45m Microbiology Lessons - Lesson 169</b><br><div>Materials: The Immortal Life of Henrietta Lacks</div><br><b>→ READ, NARRATE, &amp; DISCUSS</b><br>Read and narrate 10 pages each week. Discuss with your teacher.                                                                                                                                                                                 |                                                                                                      |
| <b>WEEK 10</b> | <b>45m Microbiology Lessons - Lesson 170</b><br><div>Materials: Illustrated Guide to Home Biology Experiments</div><br><b>→ LAB DAY</b>                                                                                                                                                                                                                                                                |                                                                                                      |

# Science: Microbiology

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## Term 3

### WEEK 10 ☐ 60m Microbiology Labs - Lesson 34

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 11 ☐ 45m Microbiology Lessons - Lesson 171

☐ Materials: I Contain Multitudes

→ READ, NARRATE, & DISCUSS

Read and narrate 8 pages each week. Discuss with others.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 11 ☐ 45m Microbiology Lessons - Lesson 172

☐ Materials: A Field Guide to Bacteria

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week.

### WEEK 11 ☐ 45m Microbiology Lessons - Lesson 173

☐ Materials: Microbe Hunters OR Growing a Revolution

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week from Microbe Hunters OR about a chapter per week from Growing a Revolution. Use questions, journal entries, charts, drawings, etc.--whatever you need to help yourself narrate and process your reading.

### WEEK 11 ☐ 45m Microbiology Lessons - Lesson 174

☐ Materials: The Immortal Life of Henrietta Lacks

→ READ, NARRATE, & DISCUSS

Read and narrate 10 pages each week. Discuss with your teacher.

### WEEK 11 ☐ 45m Microbiology Lessons - Lesson 175

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 11 ☐ 60m Microbiology Labs - Lesson 35

☐ Materials: Illustrated Guide to Home Biology Experiments

→ LAB DAY

### WEEK 12 ☐ 45m Microbiology Lessons - Lesson 176

*Exam Week*

→ EXAMS

Answer question(s) related to course.

#### • PLAN WEEKLY

- ☐ nature walk - record observations
- ☐ science free read

### WEEK 12 ☐ 45m Microbiology Lessons - Lesson 177

*Exam Week*

→ EXAMS

Answer question(s) related to course.





## Term 3

|                                                                                                                                                 |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p><b>WEEK 12</b> 45m Microbiology Lessons - Lesson 178</p> <p><i>Exam Week</i></p> <p>→ EXAMS</p> <p>Answer question(s) related to course.</p> |  |
| <p><b>WEEK 12</b> 45m Microbiology Lessons - Lesson 179</p> <p><i>Exam Week</i></p> <p>→ EXAMS</p> <p>Answer question(s) related to course.</p> |  |
| <p><b>WEEK 12</b> 45m Microbiology Lessons - Lesson 180</p> <p><i>Exam Week</i></p> <p>→ EXAMS</p> <p>Answer question(s) related to course.</p> |  |
| <p><b>WEEK 12</b> 60m Microbiology Labs - Lesson 36</p> <p><i>Exam Week</i></p> <p>→ EXAMS</p> <p>Answer question(s) related to course.</p>     |  |

# Science: Microbiology

## Examination

### Term 1

#### Microbiology Lessons

- I Contain Multitudes: Describe some of the 20th-century discoveries in the field of microbes.
- Field Guide to Bacteria: Scan your summary creation of all of the bacteria you have read about so far. Write a few paragraphs about one type.
- A Planet of Viruses: Write about the history of the study of viruses. Include some comments on COVID-19 as part of the ongoing story after the book was written; OR Microbe Hunters: Write about the discoveries of Leeuwenhoek or Spallanzani.
- The Immortal Life of Henrietta Lacks: Tell the story of Henrietta Lacks (so far) in the book.
- Describe 2 of your experiments. What conclusions did you draw from the results? What further questions did they raise? How would you go about finding the answers?
- Show your teacher/parent your nature and science notebooks. Scan a favorite page from each.

### Term 2

#### Microbiology Lessons

- I Contain Multitudes: Explain how microbes play a role in illness and in health.
- Field Guide to Bacteria: Scan your summary creation of all of the bacteria you have read about so far. Write a few paragraphs about one type.
- Micro Life: Write about a few of your favorite images. How do they help your understanding to see things up close? Include a few sketches to illustrate your points; OR Microbe Hunters: What role did Koch play in the history of microbes?
- The Immortal Life of Henrietta Lacks: What questions about bioethics have you had after reading the story of Henrietta Lacks to this point?
- Describe 2 of your experiments. What conclusions did you draw from the results? What further questions did they raise? How would you go about finding the answers?
- Show your teacher/parent your nature and science notebooks. Scan a favorite page from each.

### Term 3

#### Microbiology Lessons

- I Contain Multitudes: Describe the future of microbe research.
- Field Guide to Bacteria: Scan your summary creation of all of the bacteria you have read about so far. Write a few paragraphs about one type.
- Growing a Revolution: Why is David Montgomery optimistic about agriculture? What are ways to leave the soil better off instead of degrading it?
- The Immortal Life of Henrietta Lacks: Write 5 response statements or 5 new questions about the story after finishing The Immortal Life of Henrietta Lacks.
- Describe 2 of your experiments. What conclusions did you draw from the results? What further questions did they raise? How would you go about finding the answers?
- Show your teacher/parent your nature and science notebooks. Scan a favorite page from each.